

Zan Ahmad

Department of Applied Mathematics & Statistics
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🔗 Google Scholar 🐙 GitHub

Education

Johns Hopkins University , Whiting School of Engineering Ph.D. Candidate, Applied Mathematics & Statistics Advisors: Natalia Trayanova , Mauro Maggioni	June 2022 – May 2027 (expected)
Johns Hopkins University , Whiting School of Engineering M.S.E., Applied Mathematics & Statistics	June 2022 – May 2025
New York University , Courant Institute of Mathematical Sciences B.A., Mathematics (Honors) Minors: Chemistry, Film Production Advisors: Charles S. Peskin , Charles Puelz	Sept 2017 – Dec 2020

Research Interests

Geometric machine learning · Riemannian shape analysis · Neural operators for PDEs · Cardiovascular modeling

Research Experience

Johns Hopkins University , Institute of Computational Medicine Graduate Researcher, Computational Cardiology Lab Advisors: Natalia Trayanova , Mauro Maggioni	June 2022 – Present
Yale School of Medicine , Department of Neurology Postgraduate Researcher, Blumenfeld Lab Advisor: Hal Blumenfeld	June 2021 – May 2022
New York University , Center for Neural Science NIH Training Program in Computational Neuroscience Advisor: Charles S. Peskin	Sept 2020 – May 2021
New York University , Courant Institute of Mathematical Sciences Applied Mathematics Summer Undergraduate Research Experience (AM-SURE) Advisors: Charles Puelz , Charles S. Peskin	June 2020 – Aug 2020
New York University , Courant Institute of Mathematical Sciences Modeling & Simulation Research Training Group Advisor: Charles Puelz	Sept 2019 – May 2020
New York University , Department of Chemistry Undergraduate Researcher, DNA Nanotechnology Lab Advisors: Nadrian Seeman , Yoel P. Ohayon	Jan 2018 – Aug 2019

Publications & Preprints

[†]equal contribution. [Full list on Google Scholar.](#)

Published

1. “A Computational Pipeline for Clustering Left Atrial Appendage Morphology via Elastic Shape Analysis.”
Z. Ahmad, M. Yin, Y. Sukurdeep, N. Rotenberg, E. Kholmovski, N. Trayanova.
Computers in Biology and Medicine, 196:110905, 2025.
2. “A Comprehensive Stroke Risk Assessment by Combining Atrial Computational Fluid Dynamics Simulations and Functional Patient Data.”
A. Zingaro, **Z. Ahmad**, E. Kholmovski, K. Sakata, L. Dede’, A.K. Morris, A. Quarteroni, N. Trayanova.
Scientific Reports, 14:9515, 2024.
3. “Optimal Fenestration of the Fontan Circulation.”
Z. Ahmad, L.H. Jin, D.J. Penny, C. Rusin, C.S. Peskin, C. Puelz.
Frontiers in Physiology: Computational Physiology and Medicine, 13, 2022.

Workshop Papers

1. “Diffeomorphic Latent Neural Operators for Data-Efficient Learning of Solutions to PDEs.”
Z. Ahmad, S. Chen, M. Yin, A. Kumar, N. Charon, N. Trayanova, M. Maggioni.
AAAI 2025 Workshop on AI for Accelerating Science and Engineering, 2024.
2. “Convolutional Deep Operator Networks for Predicting Nonlinear Wave Propagation in Heterogeneous Spinal Cord Anatomy.”
A. Kumar, X. Zhi, **Z. Ahmad**, M. Yin, A. Manbachi.
AAAI 2025 Workshop on AI for Accelerating Science and Engineering, 2024.

Under Review & Preprints

1. “Predicting the *When*: Multimodal AI for Time-to-Recurrence Analysis After Atrial Fibrillation Ablation.”
M. Yin, C. Lai, R. Yadav, . . . , **Z. Ahmad**, S. Chen, . . . , N. Trayanova.
medRxiv preprint, 2026.
2. “Stimulation of the Thalamus for Arousal Restoral in Temporal Lobe Epilepsy (START) Clinical Trial: Modulation of Natural Sleep and Focal Seizures.”
T. Yadav, Z. Zhang, V. Kremen, **Z. Ahmad**, . . . , H. Blumenfeld.
medRxiv preprint, 2025.
3. “Low and High Frequency Signatures of Impaired Consciousness in Temporal Lobe Seizures.”
T. Yadav, . . . , **Z. Ahmad**, . . . , H. Blumenfeld.
bioRxiv preprint, 2025.
4. “Steady-State Analysis of Gravitational Effects on Hemodynamics.”
A. Kennard[†], **Z. Ahmad**[†], R.S. Blue, C.S. Peskin, K.M. Ong.
Submitted to Scientific Reports, 2025.
5. “Thinner Latent Spaces: Detecting Dimension and Imposing Invariance Through Conformal Autoencoders.”
G. Kevrekidis, **Z. Ahmad**, S. Villar, M. Maggioni, Y. Kevrekidis.
Submitted to SIAM Journal on Mathematics of Data Science, 2024.
6. “Graph Fourier Neural Kernels (G-FuNK): Learning Solutions of Nonlinear Diffusive Parametric PDEs on Multiple Domains.”
S. Loeffler, **Z. Ahmad**, S.Y. Ali, C. Yamamoto, D.M. Popescu, A. Yee, Y. Lal, N. Trayanova, M. Maggioni.
arXiv preprint, 2024.

In Preparation

1. “Toward a mechanistic understanding of control during simulated blood loss using compartmental models and time-frequency analysis.”
C. Puelz, B.E. Griffith, J.L. Marzuola, **Z. Ahmad**, C.S. Peskin, H.-T. Wu.
Manuscript in preparation. [[Password-protected manuscript for hiring committees](#)]

Teaching

Teaching portfolio and materials

Probability & Statistics for Life Sciences <i>Four-credit departmental course; lectures, office hours, and exam grading within a shared curriculum</i> Materials ↗	F 2026	JHU	Instructor of Record
Introduction to Scientific Programming in Python <i>Scientific libraries, numerical linear algebra, optimization, ML (neural networks, CNNs, transformers, LLMs); algorithmic-thinking and NumPy assignment</i> Assignment ↗ · Materials ↗	W 2026	JHU	Instructor of Record
Biomedical Engineering Innovation & Design	Su 2025	JHU	Instructor of Record
Biomedical Engineering Innovation & Design	Sp 2025	JHU	Instructor of Record
Introduction to Scientific Programming in Python Materials ↗	W 2025	JHU	Instructor of Record
Advanced Topics in Computational Cardiology <i>0D/1D modeling, shape analysis, manifold learning, computer vision, neural operators; upper-level seminar. Heart-rate feedback-control project; finite-difference Python tutorial</i> Feedback control ↗ · Python tutorial ↗ · Materials ↗	F 2024	JHU	Creator; Instructor of Record
Mathematical Modeling & Computer Simulation <i>First-principles biological modeling; course projects served as launching points for undergraduate research mentorship</i> Materials ↗	F 2024	JHU	Creator; Instructor of Record
Biomedical Engineering Innovation & Design	F 2024	JHU	Instructor of Record
Biomedical Engineering Innovation & Design	Su 2024	JHU	Instructor of Record
Mathematical Cardiology <i>First-principles cardiovascular modeling as an accessible on-ramp to research in mathematical biology and cardiovascular modeling</i> Materials ↗	F 2023	JHU	Creator; Instructor of Record
Biomedical Engineering Innovation & Design	Su 2023	JHU	Instructor of Record
Computational Neuroscience: Channels & Networks <i>Developed MATLAB tutorials on neuronal action potential models; students used these as a foundation for final course projects. [tutorial videos]</i> Materials ↗	Sp 2021	NYU	Guest Lecturer
Linear Algebra	F 2020	NYU	Grader
Calculus I	Sp 2020	NYU	Grader
Calculus I	F 2019	NYU	Grader
Thermodynamics; Organic Chemistry I & II <i>F 2019: Thermodynamics and Organic Chemistry II; Sp 2019: Organic Chemistry I</i>	Sp–F 2019	NYU	Lead Chemistry Tutor
Physics <i>East Harlem Tutorial Program: nonprofit providing free tutoring and mentorship to K–12 students in East Harlem, NY</i>	2018–19	EHTP	Volunteer Tutor

Mentoring

Shiyi Chen · M.S. student, Mechanical Engineering, Johns Hopkins University Apr 2024 – Aug 2025
 Thesis: [Learning PDE solution operators via diffeomorphic mappings: applications in fluid dynamics](#).
 Next position: Ph.D. student, Biomedical Engineering, Yale University.

Ayush Mandal · Undergraduate, Applied Mathematics & Computational Neuroscience, JHU Topics: Feedback control of heart rate in zero-dimensional circulation models; modeling neuronal dynamics.	Sept 2023 – Nov 2025
Noam Rottenberg · Undergraduate, Engineering, Johns Hopkins University Topics: Automatic left atrial segmentation and preprocessing for statistical shape analysis. <i>Next position: Medical student, Yale School of Medicine.</i>	Aug 2023 – Jan 2024

Industry Experience

Intuit Inc. , AI/Data Science Department · San Diego, CA AI Science Intern, Consumer Group Analytics and Decision Science Advisor: Dr. Joseph Powers	June 2025 – Sept 2025
Virgin Galactic · Remote Computational Research Intern Advisor: Dr. Karen Ong	Jan 2023 – Aug 2023

Grants, Awards & Honors

American Heart Association Predoctoral Fellowship · \$67,388 <i>Quantifying Stroke Risk in Atrial Fibrillation with Computational Fluid Dynamics and Machine Learning.</i>	July 2024 – June 2026
IEEE EMBS Student Fellowship · \$2,000 One of 14 selected from 900 applicants. Full scholarship for AI in Medicine & Healthcare workshop. Chania, Crete, Greece.	June 2024
IDIES Poster Competition — 1st Place · \$100 <i>Random Convolutional Features and Patch-Based Learning for Multitask Image Classification.</i>	Oct 2023
NIH Cardiovascular Epidemiology Training Grant (T32HL007024) · \$27,400 Fellowship for PhD students with research focused on cardiovascular disease.	July 2023 – June 2024
Acheson J. Duncan Fund Travel Award · \$3,150 Fund for the Advancement of Research in Statistics.	April 2023
NIH Computational Medicine Training Grant (T32GM119998) · \$25,836 Fellowship for PhD students at the Institute of Computational Medicine, JHU.	June 2022 – May 2023
NIH Computational Neuroscience Training Grant (R90DA043849) · \$13,336 Awarded to 6 of 50 applicants. <i>Short time-scale dynamics in a computational model of signal propagation in a cerebellar neuronal network.</i>	Aug 2020 – May 2021
NSF Modeling & Simulation RTG (DMS-1646339) · \$6,250 AM-SURE at NYU Courant Institute. Three undergraduates selected from 60 applicants.	May – Aug 2020
Wilfred L. & Ruth S.F. Peltz Research Scholar Award · \$2,750 Awarded for NYU DURF proposal: <i>Optimally fenestrated Fontan circulation.</i>	May 2020
Dean's Undergraduate Research Fund Grant (DURF) · \$1,000 <i>Optimal fenestration of Fontan circulation.</i> NYU College of Arts and Sciences.	May 2020
Dean's Undergraduate Research Fund Training Grant · \$500 <i>Self-assembly of A-form 3D DNA crystals of 2'-fluoro DNA.</i>	May 2018

Seminars & Invited Talks

1. “A Scalable Approach to Riemannian Geometric Statistics for Cardiovascular Shape Modeling.”
Mostly Biomathematics Lunchtime Seminar, April 2026.
2. “Neural Operators for Learning PDEs on Geometrically Varying Data.”
JHU Applied Mathematics & Statistics Department Seminar, April 2024 & December 2024.
3. “Hemodynamic Indicators of Stroke Risk Uncovered via Personalized Fluid Dynamics Simulations.”
JHU Applied Mathematics & Statistics Department Seminar, September 2023.

4. “Steady-State Analysis of Gravitational Effects on Hemodynamics.”
JHU Applied Mathematics & Statistics Department Seminar, February 2023.
5. “Pulsatile Model for the Fontan Circulation and a Feedback Control Method for Heart Rate.”
JHU Applied Mathematics & Statistics Department Seminar, November 2022.
6. “Short Time-Scale Dynamics in a Computational Model of Signal Propagation in a Cerebellar Neural Network.”
NYU Training Program for Computational Neuroscience Conference, June 2021.
7. “Using Mathematics to Study and Optimize Congenital Heart Disorders.”
NYU College of Arts and Sciences Undergraduate Research Conference, May 2021.
8. “Feedback Control of Blood Pressure and Heart Rate in a Pulsatile Cardiovascular Model.”
NYU Courant Institute Modeling and Simulation Research Training Group Seminar, March 2021.
9. “Mathematical Model for an Optimally Fenestrated Fontan Circulation.”
NYU Courant Institute Applied Mathematics Research Presentations, October 2020.

Conference Presentations

1. “AI-Based Shape Analysis Correlates Left Atrial Appendage Morphology with Stroke Risk in Patients with Atrial Fibrillation.”
Heart Rhythm Society (HRS), San Diego, CA, April 2025.
2. “Multimodal Artificial Intelligence for Predicting Atrial Fibrillation Recurrence Using Pre-Procedural Data.”
Heart Rhythm Society (HRS), San Diego, CA, April 2025.
3. “Mathematical Cardiology: Operator Learning, Shape Analysis, and Computational Fluid Dynamics.”
IEEE EMBS Leadership Academy Workshop on AI/Data Science in Medicine and Healthcare, Chania, Crete, Greece, June 2024.
4. “Random Convolutional Features and Patch-Based Learning for Multitask Image Classification.”
IDIES Annual Research Symposium, Baltimore, MD, October 2023. **1st Place Best Poster Award.** [\[poster\]](#)
5. “Random Convolutional Features and Patch-Based Learning for Multitask Image Classification.”
Johns Hopkins University AI-X Symposium, Baltimore, MD, September 2023.
6. “Non-Pulsatile ODE Model of Circulation Response to Hypergravity.”
SIAM Conference on Control Theory, Philadelphia, PA, July 2023.
7. “Modeling Feedback Control Response to Hemodynamic Perturbations in a Pulsatile Circulation Model.”
SIAM Conference on Control Theory, Philadelphia, PA, July 2023.
8. “Hemodynamics and Oxygen Model for the Fontan Circulation with Feedback Control in Variable Heart Rate.”
ECCOMAS Minisymposium on Multiphysics and Multiscale Simulations in Cardiovascular Applications, Porto, Portugal, June 2023.
9. “A Dynamic Mathematical Model for the Prediction of Cardiovascular Response to Hypergravity.”
American Society of Aerospace Medicine (ASMA), New Orleans, LA, May 2023.
10. “Stroke Risk Is Identified by Slow Blood Flow in the Left Atrium.”
Heart Rhythm Society (HRS), New Orleans, LA, May 2023.
11. “Modeling the Interplay Between Blood Flow and Oxygen Transport in the Defective Heart.”
SIAM Conference on Dynamical Systems, New Orleans, LA, May 2023.
12. “Hemodynamic Indicators of Stroke Risk Uncovered by Personalized Fluid Dynamics Models.”
Cardiac Physiome Workshop, Irvine, CA, May 2023.
13. “A Dynamic Model for the Control of Cardiovascular Response to Spaceflight.”
International Conference of Aerospace Medicine, Paris, France, September 2022.
14. “Simulation of Circulation Response to Gravitational Changes During Spaceflight.”
SIAM Conference on Life Sciences, Pittsburgh, PA, July 2022.
15. “Optimal Fenestration of the Fontan Circulation.”
SIAM Annual Meeting, Pittsburgh, PA, July 2022. [\[report\]](#)
16. “Mathematical Modeling of Cardiac Circulation Under Hypergravity.”
American Society of Aerospace Medicine (ASMA), Reno, NV, May 2022.
17. “Increased Intracranial EEG Power and Duration in Temporal Lobe Seizures with Impaired Consciousness.”
American Epilepsy Society, Chicago, IL, December 2021.

18. “Mechanisms of Impaired Consciousness in Medial Temporal Lobe Seizures Investigated with Intracranial EEG.”
Society for Neuroscience (SfN), Chicago, IL, November 2021.
19. “Mathematical Model for an Optimally Fenestrated Fontan Circulation.”
Undergraduate Mathematics Symposium, University of Illinois at Chicago, Chicago, IL, December 2021.

Service & Leadership

- Founder and President, **JHU SIAM Student Chapter**. Lead organizer of inaugural SIAM DMV Area Applied Math Conference, April 2025 — full-day event with UMDCP, UMBC, GWU, GU, GMU, and JHU, over 100 attendees.
- Editor-in-Chief, **NYU Medical Dialogue Review**.
- Volunteer Physics Tutor, **East Harlem Tutorial Program (EHTP)** — January 2019 Volunteer of the Month.
- Volunteer Tutor, **The Perfect Tutor** (NYU non-profit) — Lead Math Tutor.

Media Publications

1. [“Circulation Models Assess the Impacts of Congenital Heart Defects and Hypergravity.”](#)
Jillian Kunze, *SIAM News*, August 2023. *Coverage of SIAM Control Theory Conference presentation.*
2. [“Medicine and Biology Through the Lens of Mathematics.”](#)
Z. Ahmad, *NYU Medical Dialogue Review*, Vol. XIV, Issue 2, pp. 44–47, April 2020.